Immunosenescence and cancer: mechanisms and therapeutic targets

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ABSTRACT

Immunosenescence contributes to tumor progression. This review summarizes current knowledge on immune aging and cancer immunotherapy.

Keywords: immunosenescence; cancer; immune aging

1 Introduction

Although our understanding of immunosenescence has steadily progressed over the past few decades and many studies on age-related immune decline have laid the foundation for identifying intervention methods [10, 21-23], the interactions between senescence-related changes and different components of the immune system remain unclear.

Here, we discuss the most relevant strategies for current research on immunosenescence, focusing on its characteristics and on the various types of immune cells during aging, also highlighting the pivotal role of immunosenescence in tumor progression and immunotherapy. Moreover, we also discuss the interaction between cancer cells and the aging TME, and address how the latter drives tumor progression.

According to United Nations projections, the number of people over 60 years of age will increase from 962 million in 2017 to 1.4 billion in 2030 and 2.1 billion by 2050. This would reflect a projected increase from 12.8 % (2017) to 21.3 % of the world's population [1].

2 T cells and aging

P53 can cause cellular senescence [42] by regulating the cell cycle via inhibition of cyclin-dependent kinase (CDK) 4 and CDK6 [5, 43] (Fig. 2).
Another hallmark of senescent
T cells is telomere shortening [21, 44] (Fig. 2), which is caused by the continuous replication of T cells.

This process is considered to be a manifestation of immunosenescence. Therefore, understanding how to restore thymus function and the production of T cells provides an effective strategy for immunosenescence intervention.

3 Conclusion

Immunosenescence is a key driver of tumor progression.

Abbreviations

cAMP: Cyclic adenosine monophosphate; CAR-T: Chimeric antigen receptor; CTLA-4: Cytotoxic T-lymphocyte-associated protein-4; TME: Tumor microenvironment; Tregs: Regulatory T cells.

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Competing interests

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Fig. 2 Hallmarks of senescent T cells including cell cycle arrest and telomere shortening.